(1)

1) the MRT of Consumption is how much future Consumption must be given up in $t^{*+1}$ for 1 unit of extra Consumption ~~tew~~ in $t^*$.

a. at $t^*$ Price $= P$

$t^*_{+1}$ Price $= (1+\theta)P$

But any given up consumption today gives $1+r$ more consumption tomorrow so:

$$MRT = \frac{P(1+r)}{(1+\theta)P} = \frac{1+r}{1+\theta}$$

B. $$MRT = \frac{P(1+\theta)(1+r)}{P(1+\theta)} = 1+r$$

II) without suspension:

$$u'(c_{t^*}) = \frac{1+r}{1+\delta} E\left[u'(c_{t^*+1})\right]$$

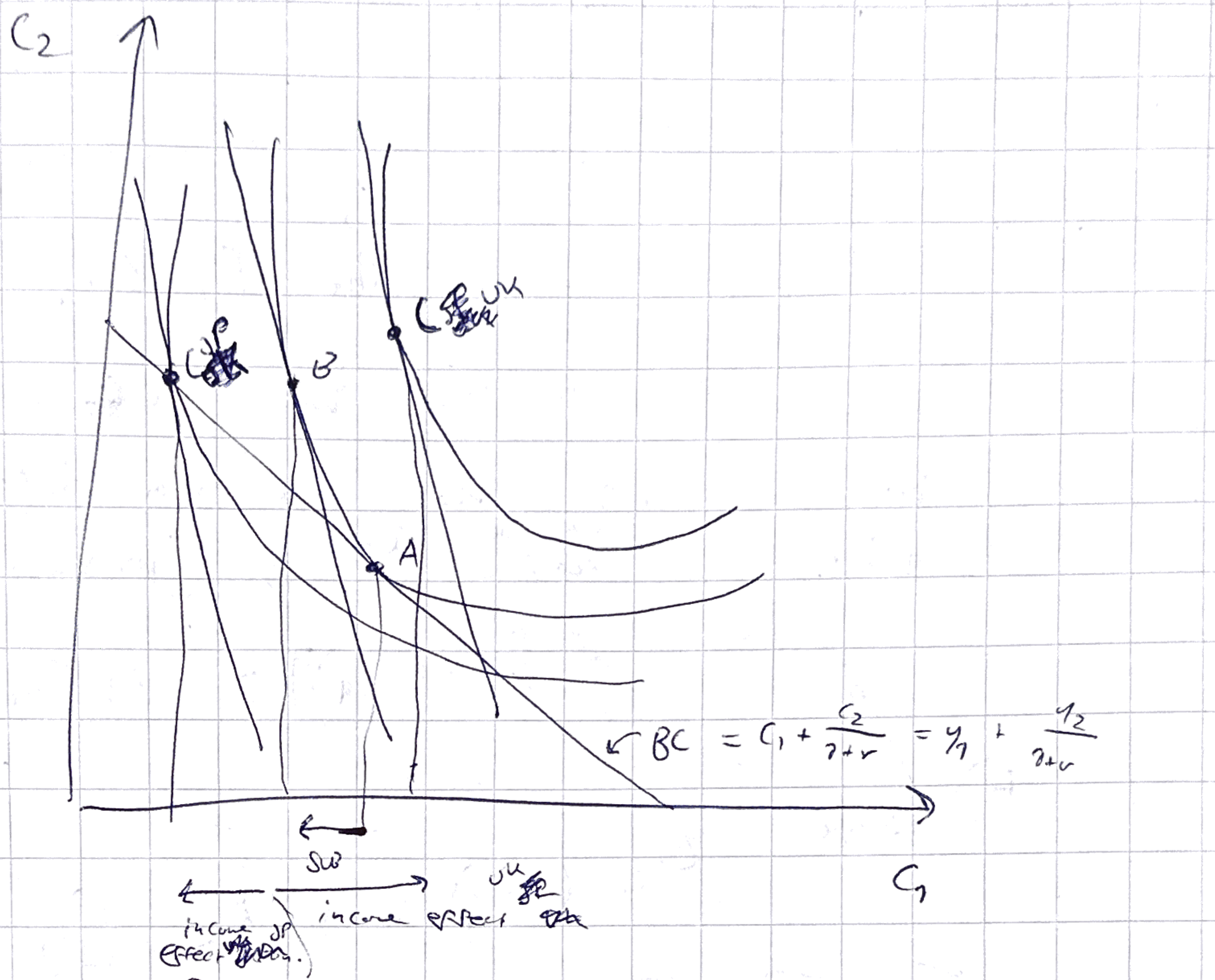
However, since consumption in $t^*+1$ is $(1+\theta)$ more expensive:

$$u'(c_{t^*}) = \frac{1+r}{(1+\delta)(1+\theta)} E\left[u'(c_{t^*+1})\right]$$

III) $u'(c_t) = \frac{1+r}{(1+\delta)(1+\theta)} E[u'(c_{t+1})]$

- the tax makes it better to increase consumption in $t^{*}$ and reduce in $t^{*}+1$ since it is ~~more~~ cheaper in $t^{*}$.
- ~~EIS~~ $r$ & $\delta$, of course, with opposite effect.
- EIS, the higher the EIS, the more willingness to ~~be~~ change consumption so $C_t$ will be higher

2




a fall in r makes the BC steeper because more of $C_1$ has to given up for $C_2$. But because in Japan they are net savers, the income effect move the BC ~~higher~~ left but in the UK ~~left~~ right since they have to pay more interest.

(3) I) Hall tries to prove from

$U'(c_t) = E[u'(c_{t+1})]$ that $c_t = E[c_{t+1}]$

However, that requires that $E[u'(c_{t+1})] = u'(E[c_{t+1}])$

Which is only the case when $u'$ is linear.

This is not the case with $\ln(c_t)$

Which is ~~convex~~ concave So By Jensen's

inequality $E[u'(c_{t+1})] > u'(E[c_{t+1}])$

II) Since $E[u'(c_{t+1})] > u'(E[c_{t+1}])$

~~So uncertainty raises expected marginal utility~~

~~relative to~~ the agent is risk averse so will do precautionary savings creating excess smoothness.

~~III) So~~

(4)

I) the Euler equation is

$$u'(c_t) = \beta(1+r) E[u'(c_{t+1})]$$

assuming $\beta(1+r) = 1$

$$u'(c_t) = E[u'(c_{t+1})]$$

assuming a utility function with linear $u'(c_t)$, such as a quadratic:

$$c_t = E(c_{t+1})$$

So therefore, consumption follows a random walk:

$$c_{t+1} = c_t + \epsilon_{t+1}, \quad E[\epsilon_{t+1}] = 0$$

II) $A_{t+1} = (1+r)A_t + y_t - c_t$

$$\frac{A_{t+1}}{(1+r)^t} = \frac{(1+r)A_t}{(1+r)^t} + \frac{y_t - c_t}{(1+r)^t} = \frac{A_t}{(1+r)^{t-1}} + \frac{y_t - c_t}{(1+r)^t}$$

Now if we sum all of the assets up to $t$ together, we get

$$\cancel{A_1} + \cancel{\frac{A_2}{(1+r)}} + \dots + \cancel{\frac{A_{t-1}}{(1+r)^{t-2}}} + \frac{A_t}{(1+r)^{t-1}} = (1+r)A_0 + y_0 - c_0$$

$$+ \cancel{A_1} + \frac{y_1 - c_1}{\dots} + \dots + \cancel{A_{t-1}} + \frac{y_{t-1} - c_{t-1}}{\dots}$$

Letting $t \to +\infty$ and assuming the no Ponzi game condition:

$$\lim_{t \to +\infty} E\left[\left(\frac{1}{1+r}\right)^t A_{t+1}\right] = 0$$

$$E\left[\sum_{t=0}^{+\infty} \frac{c_t}{(1+r)^t}\right] = (1+r)A_0 + E\left[\sum_{t=0}^{+\infty} \frac{y_t}{(1+r)^t}\right]$$

$$= (1+r)A_0 + \sum_{t=0}^{+\infty} \frac{E[y_t]}{(1+r)^t}$$

$E(y_t) = y^*$

so

$$= (1+r)A_0 + \frac{1+r}{r} y^*$$

III) By the random walk model

$$E(c_t) = c_0$$

Using the intertemporal budget constraint:

$$E\left[\sum_{t=0}^{+\infty} \frac{c_t}{(1+r)^t}\right] = (1+r)A_0 + E\left[\sum_{t=0}^{+\infty} \frac{y_t}{(1+r)^t}\right]$$

$$\frac{1+r}{r} c_0 = (1+r)A_0 + E\left[\sum_{t=0}^{+\infty} \frac{y_t}{(1+r)^t}\right]$$

$$c_0 = rA_0 + \frac{r}{1+r} E\left[\sum_{t=0}^{+\infty} \frac{y_t}{(1+r)^t}\right]$$

Since $y_0$ is known:

$$E\left[\sum_{t=0}^{+\infty} \frac{y_t}{(1+r)^t}\right] = y_0 + E\left[\sum_{t=1}^{+\infty} \frac{y_t}{(1+r)^t}\right]$$

$$= y_0 + \sum_{t=1}^{+\infty} \frac{y^*}{(1+r)^t}$$

$$= y_0 + \sum_{t=0}^{+\infty} \frac{y^*}{(1+r)^t} - y^*$$

$$= y_0 + \frac{1+r}{r} y^* - y^*$$

$$= y^0 + \frac{1}{r} y^*$$

So $C_0 = rA_0 + \frac{r}{1+r}\left(y_0 + \frac{1}{r} y^*\right)$

So when $y_0 = y_L$ or $y_0 = y_H$, $C_0$ falls/rises but less than one-for-one